Structural Equation Modeling with Onyx

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Invalid Date

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Preface

This is a Quarto book.

To learn more about Quarto books visit https://quarto.org/docs/books.

1 Introduction

This is a self-paced workshop for Onyx, a graphical user-interface for structural equation modeling.

2 LGCM

3 Untitled

3.1 Model Fit

- χ^2 is the fundamental model fit measure as discrepancy between model-implied (co)variances and means and observed (co)variances and means
- Significance test of close fit is usually not recommended as it has very high power; instead we usually favor effect-size-type metrics
- CFI: discrepancy between an independence model and the hypothesized model; Q: Is there enough information in the covariances such that the model is supported?
- RMSEA: discrepancy between the saturated model and the hypothesized model while acknowledging sample size and model complexity; Q: Are all those free parameters warranted?
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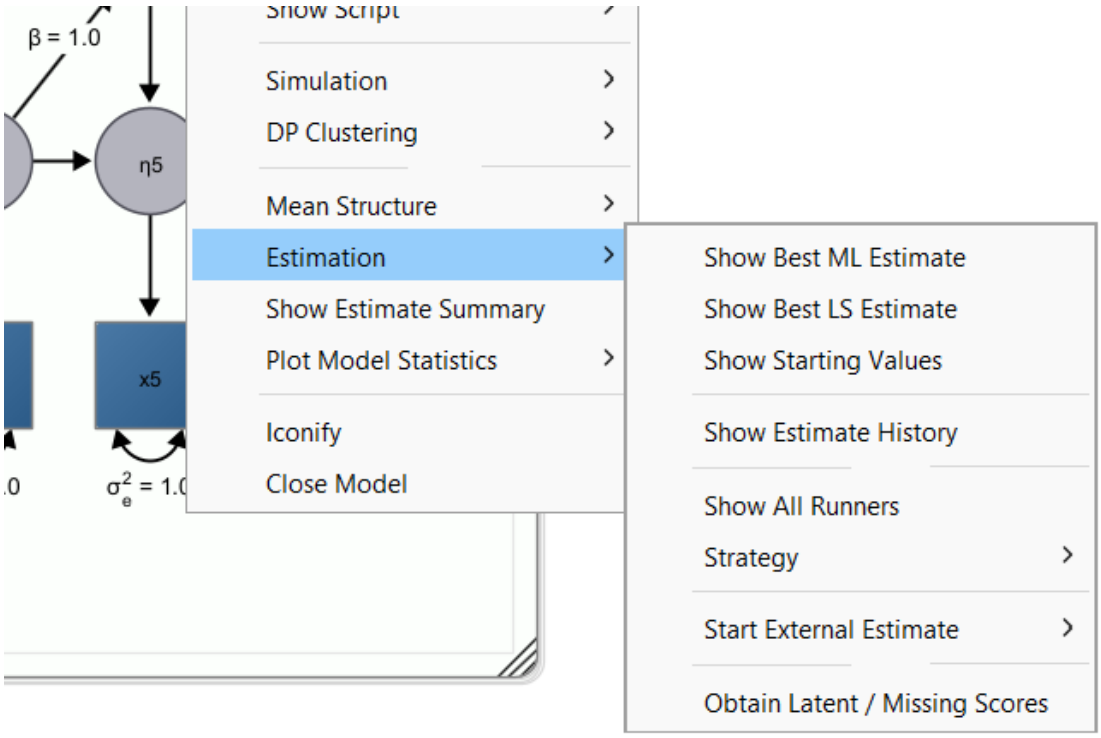
3.2 Model Fit Plots

As of Onyx X, there are two new plotting features to obtain visual information about model fit.



uarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see https://quarto.org.

3.3 Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

```
[1] 2
```

You can add options to executable code like this

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

4 Simulate and Plot

> Goal: Learn about simulation and plotting of data

4.1 Simulation to Understand Model Parameters

We want to answer this question:

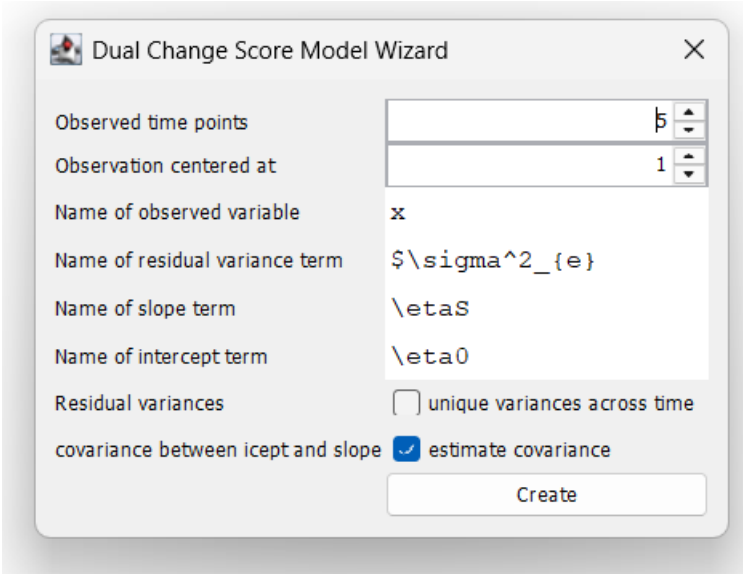
In a Dual Change Score Model, what effects do parameters have on the model-implied trajectories?

Steps:

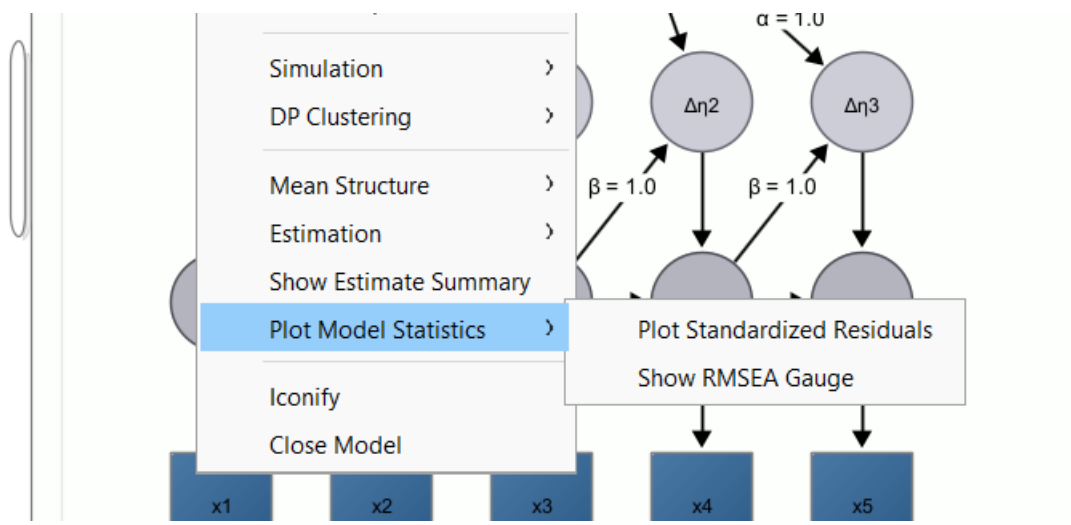
1. Create a Dual Change Score Model
2. Simulate Longitudinal Data
3. Plot Longitudinal Data
4. Change Model (Population) Parameters and Repeat from 2

4.2 Dual Change Score Model

Use the DCSM-Wizard to create a dual change score model. Keep all values at their defaults. This creates a model with five time points and the observerd time points will be called x1, x2, ..., x5.

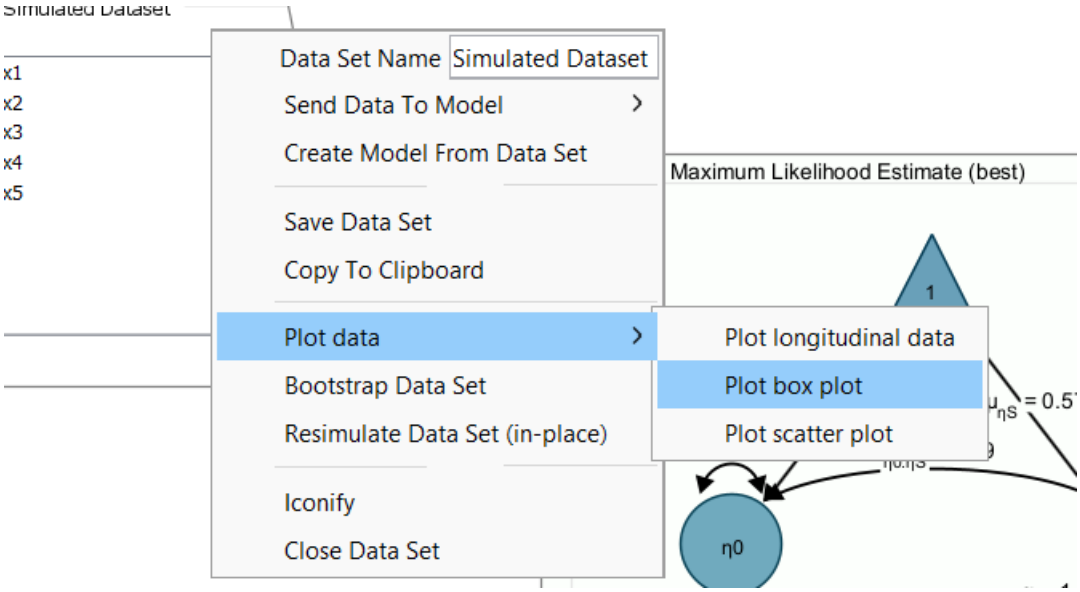


4.3 Simulation

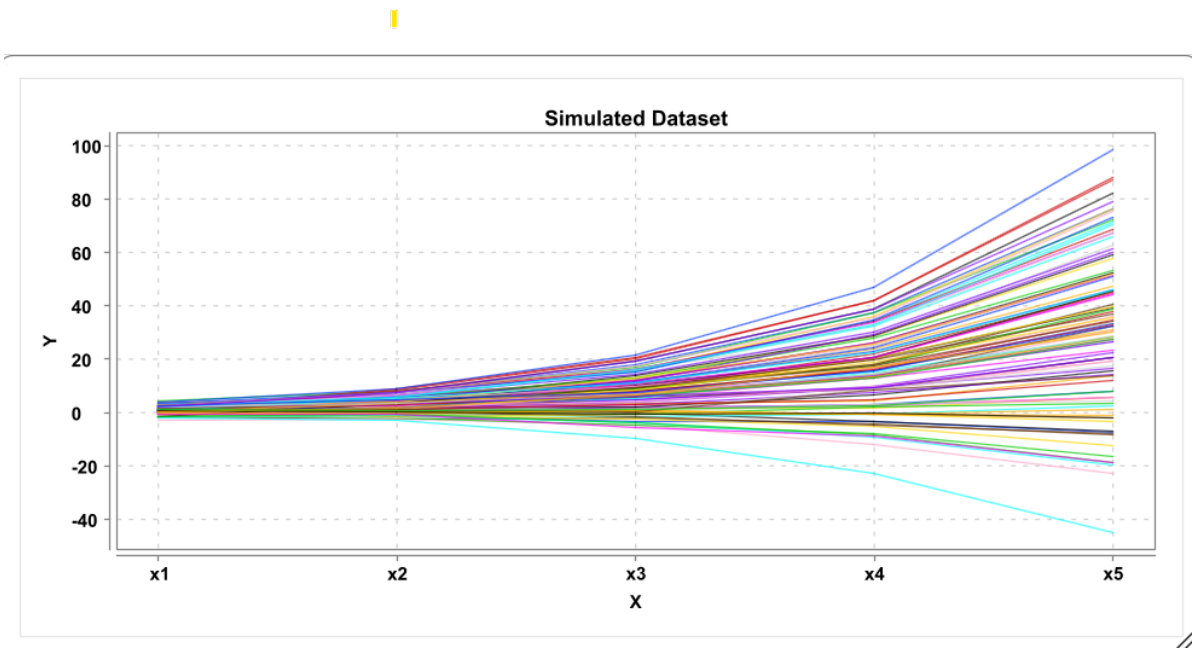


4.4 Plot

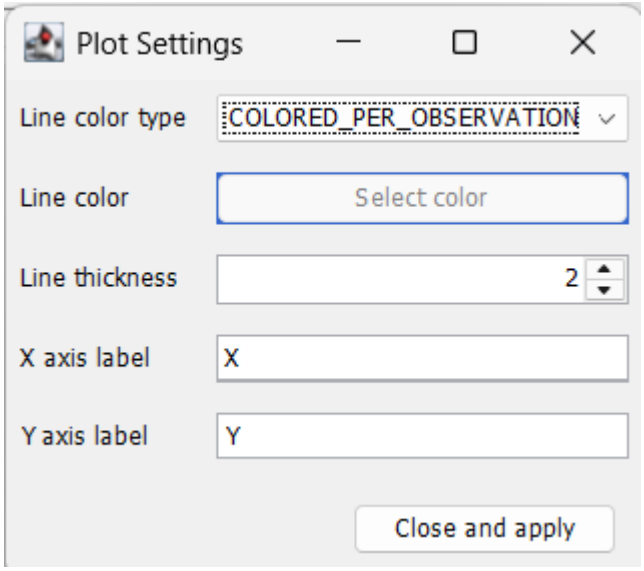
Next, right-click on the simulated dataset, choose "Plot data" and select "Plot longitudinal data".



4.5 Longitudinal Plot



4.6 Customize the Longitudinal Plot



5 Summary

In summary, this book has no content whatsoever.